

Wireless Network Design Project

1. Project Overview

This project aims to evaluate students' ability to design and implement a complete wireless network infrastructure using Cisco Packet Tracer.

Students are required to simulate a real-world company environment consisting of a multifloor office building and an additional remote department. The network must ensure full connectivity, proper configuration, and secure wireless communication between all devices.

2. Company Description

The company consists of two main parts:

- four-floor main office building
- separate department located outside the main building

All networks must be interconnected through a central router to allow seamless communication.

3. Network Design

3.1 Main Office Building

The building consists of four floors, each representing a different department. Each floor must be equipped with an Access Point to provide wireless connectivity.

All devices are connected wirelessly to the department's wireless router.

Technical Requirements assign valid IP addresses to

all devices based on:

* Main Building: `192.168.20.0/24`

* Separate Department: `192.168.10.0/24`

Wireless Configuration

- * Configure a unique SSID for each floor
- * Secure all wireless networks using WPA2
- * Ensure successful wireless connectivity